MOMENT EXPONENTS FOR GAUSSIAN DIRECTED POLYMERS IN THREE DIMENSIONS

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ABSTRACT. For the directed polymer based on simple random walk in $\mathbb{Z}^3$ with iid standard Gaussian disorder, let $p^*(\beta)$ be the supremum of the orders for which the normalized partition function has moments bounded uniformly in the polymer length. We prove that $\beta \mapsto p^*(\beta)$ is a continuous strictly decreasing bijection from $(0, \beta_c]$ onto $[5/3, \infty)$, where β_c is the weak-disorder critical point. The main step is local stability of every strictly subcritical moment above order two under an increase of β. A concave power of a terminal polymer moment has a one-step deficit proportional to the endpoint overlap; Gaussian differentiation bounds the perturbation by the same overlap. This yields a moment bound for all lengths. Gaussian hypercontractivity and the established critical and one-sided continuity results then give the full statement.

1. INTRODUCTION

A directed polymer in a random environment weights random-walk trajectories according to the disorder they encounter. Its normalized partition function is a positive mean-one martingale. The weak-disorder phase is characterized by a positive martingale limit, but its fluctuations need not have a bounded second moment. The threshold for higher moments gives a finer description of this phase; see [1, Section 3.3.2] for the moment-spectrum question and its relation to the hierarchy of L^p critical points.

We consider simple random walk $S = (S_n)_{n\geq 0}$ from zero on $\mathbb{Z}^3$, with each of its six nearest-neighbor steps having probability $1/6$. Let $(\omega_{n,x})_{n\geq 1, x\in\mathbb{Z}^3}$ be iid standard real Gaussians independent of the walk. For inverse temperature $\beta > 0$, define

$$W_N^\beta = E_0 \exp\left\{\sum_{n=1}^{N} (\beta\omega_{n,S_n} - \beta^2/2)\right\}, \qquad N \geq 0, \tag{1.1}$$

where E_0 denotes walk expectation and $\mathbb{E}$ denotes disorder expectation. We set $W_0^\beta = 1$ and write W_∞^β for the almost sure martingale limit. The moment exponent and the weak-disorder critical point are

$$p^*(\beta) = \sup\{u \geq 1 : \sup_{N\geq 0} \mathbb{E}[(W_N^\beta)^u] < \infty\}, \tag{1.2}$$

$$\beta_c = \sup\{\beta > 0 : \mathbb{P}(W_\infty^\beta > 0) = 1\}. \tag{1.3}$$

All moment orders are real. The supremum in (1.2) does not assert boundedness of the moment at its endpoint.

Several ingredients of the spectrum are known for much more general disorder. Weak disorder holds below the critical point and also at criticality for the simple walk in dimension at least three; in dimension three the critical exponent is $5/3$ [6, Theorem 2.1 and Corollary 2.5]. Junk and Lacoin prove left continuity wherever weak disorder holds [5, Corollary 2.11(i)]. Junk proves right continuity when $5/3 < p^*(\beta) \leq 2$ [4, Theorem 1.2(i)]. The strict-monotonicity statement

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in part (ii) of that theorem assumes finite-support disorder and therefore does not apply to the Gaussian law. Gaussian hypercontractivity provides a direct comparison between different inverse temperatures, as we recall in Section 2. To obtain continuity of the whole Gaussian spectrum we prove upward stability of moment bounds in the range $p^*(\beta) > 2$.

Theorem 1.1. *For the Gaussian model (1.1), $0 < \beta_c < \infty$, and*

$$(0, \beta_c] \ni \beta \longmapsto p^*(\beta) \in [5/3, \infty)$$

is a continuous strictly decreasing bijection. Continuity at β_c is from the left.

This proves the Gaussian three-dimensional instance of the continuity and strict-monotonicity conjecture stated in [1, Conjecture 3.32]. It describes every finite real value of the moment exponent on the original weak-disorder interval. It does not settle boundedness at the threshold moment order.

The stability argument uses the limiting partition functions started at different sites in a common future environment. Their joint law gives a harmonic identity for the terminal p moment of any initial probability vector. Taking its power $s/p < 1$ creates a concavity deficit. Positive association and a Gaussian change of measure bound that deficit from below by a constant times the squared ℓ^2 norm of the predicted endpoint law. The derivative with respect to the disorder variance has the same factor. A small upward change of variance is therefore absorbed uniformly, even for very spread-out endpoint laws. Iterating the resulting inequality gives the s moment bound for every polymer length. The strict gaps $2 < s < p < q < p^*(\beta)$ provide both the differentiability and the density integrability needed in this argument.

Section 2 records the precise existing continuity results and proves the Gaussian comparison. Section 3 establishes local moment stability and completes the proof of Theorem 1.1. Throughout, expectations without a walk subscript refer to the disorder; additional subscripts indicate which independent variables are averaged.

2. Gaussian comparison and continuity facts

Write $W_N^{\beta,x}$ for the partition function started at x, and put $W_0^{\beta,x} = 1$. With respect to the disorder filtration $\mathcal{F}_N = \sigma(\omega_{n,x} : n \le N)$, these are positive mean-one martingales. Indeed, averaging the fresh disorder slice gives one. For every real $u \ge 1$, Jensen's inequality over the walk paths gives

$$\mathbb{E}[(W_N^{\beta,x})^u] \le \exp\{N(u^2 - u)\beta^2/2\}. \tag{2.1}$$

The nonnegative martingale convergence theorem therefore defines $W_\infty^{\beta,x}$ almost surely; see, for example, [3, Theorem 4.2.12]. We will repeatedly use the elementary consequence of the definition of p^*: every $1 \le u < p^*(\beta)$ is a uniformly bounded moment order. To see this, choose a bounded order $v > u$ and apply Hölder's inequality. No assertion at $u = p^*(\beta)$ is needed.

There is a positive weak-disorder interval and a parameter $\beta_\ell > 0$ at which the martingale is bounded in L^2 [4, Theorem A]. In particular, $\beta_c > 0$. The opposite bound is elementary in the Gaussian case. For $0 < \theta < 1$, summing the 6^N path contributions and using subadditivity yields

$$\mathbb{E}[(W_N^{\beta})^\theta] \le \exp\{N(1 - \theta)(\log 6 - \theta\beta^2/2)\}.$$

If $\beta^2 > 2 \log 6$, choose $\theta < 1$ with $\theta\beta^2/2 > \log 6$. Fatou's lemma then gives $W_\infty^\beta = 0$ almost surely. Thus

$$0 < \beta_c \le \sqrt{2 \log 6} < \infty. \tag{2.2}$$

Similarly, for every $u > 1$, the pointwise inequality $(\sum_i a_i)^u \ge \sum_i a_i^u$ gives

$$\mathbb{E}[(W_N^{\beta})^u] \ge \exp\{N(u - 1)(u\beta^2/2 - \log 6)\}. \tag{2.3}$$

Hence $p^*(\beta)$ is finite for every $\beta > 0$; if $p^*(\beta) > 1$, it is at most $2\log 6/\beta^2$.

Let $\beta_2 = \sup\{\beta > 0 : \sup_N \mathbb{E}[(W_N^\beta)^2] < \infty\}$ denote the L^2 critical point. We use the following established results, with their original disorder assumptions specialized to standard Gaussians:

(1) Weak disorder holds for $0 < \beta \le \beta_c$, and

$$p^*(\beta_c) = \frac{5}{3}. \tag{2.4}$$

The phase intervals are stated in [4, Theorem A]. Weak disorder at criticality follows from [6, Theorem 2.1 and the discussion after Remark 2.2]; its simple-walk discussion includes the strict inequality $\beta_c > \beta_2$ in dimension three. The value in (2.4) is [6, Corollary 2.5].

(2) At every parameter where weak disorder holds,

$$\lim_{b\uparrow\beta} p^*(b) = p^*(\beta) \tag{2.5}$$

by [5, Corollary 2.11(i)].

(3) If $5/3 < p^*(\beta) \le 2$, then p^* is right continuous at β by [4, Theorem 1.2(i)].

The Gaussian law has all real exponential moments, so the unbounded-disorder results cited here apply. In [5], the threshold is written as the infimum of the orders whose uniform moments are infinite. This is the same number as (1.2): Hölder makes the bounded orders a downward interval, and (2.3) supplies finite orders outside it. Neither convention determines whether its endpoint belongs to that interval.

Gaussian hypercontractivity supplies a quantitative comparison between temperatures. For a function of a finite standard Gaussian vector, define

$$T_\rho f(z) = \mathbb{E}_G f(\rho z + \sqrt{1-\rho^2}\, G), \qquad 0 \le \rho \le 1,$$

where G is an independent standard Gaussian vector of the same dimension. Nelson's inequality states

$$\|T_\rho f\|_v \le \|f\|_u, \qquad 1 < u \le v < \infty, \quad \rho \le \sqrt{\frac{u-1}{v-1}}. \tag{2.6}$$

See [7, Theorem 1.1 in the author-generated copy]. In Ornstein–Uhlenbeck time a, this is the semigroup with $\rho = e^{-a}$; the Mehler formula and the corresponding exponent $v = 1 + (u-1)e^{2a}$ are given in [2, Lemma 1.8 and Theorem 2.1]. The constant is independent of the dimension.

Proposition 2.1. *For $0 < b < \beta$ with $p^*(\beta) > 1$,*

$$p^*(b) \ge 1 + \frac{\beta^2}{b^2}(p^*(\beta) - 1). \tag{2.7}$$

Consequently, p^* is strictly decreasing on $(0, \beta_c]$ and $\lim_{b\downarrow 0} p^*(b) = \infty$.

Proof. For fixed N, only finitely many disorder coordinates enter W_N^β. Every path uses N distinct coordinates, since their time indices differ. Integrating the added Gaussian vector in T_ρ therefore gives

$$T_\rho W_N^\beta = W_N^{\rho\beta}. \tag{2.8}$$

More explicitly, the integration contributes $\exp\{N\beta^2(1-\rho^2)/2\}$, changing the compensator from $N\beta^2/2$ to $N(\rho\beta)^2/2$ in each path term. The path sum is finite. Jensen's inequality for T_ρ, followed by Gaussian invariance, first shows that p^* is nonincreasing.

Now take $1 < u < p^*(\beta)$ and set $\rho = b/\beta$, $v = 1 + (\beta/b)^2(u-1)$. Equations (2.6) and (2.8) transfer the uniform u moment at β to a uniform v moment at b. Letting $u \uparrow p^*(\beta)$ proves (2.7).

This passes to a supremum of admissible orders, rather than asserting a bounded moment at that supremum.

Monotonicity and (2.4) give $p^*(\beta) \geq 5/3$ for $\beta \leq \beta_c$. Since $\beta^2/b^2 > 1$, (2.7) is then strictly larger than $p^*(\beta)$. Finally, the actually bounded second moment at β_ℓ and (2.6) give $p^*(b) \geq 1 + \beta_\ell^2/b^2$ for $0 < b < \beta_\ell$. □

3. Stability of moments above two

The remaining step is an upward perturbation of the inverse temperature. The moment loss in the statement below leaves room for strict concavity and an integrable change of measure.

Proposition 3.1. *Let $\beta_0 > 0$ and $2 < s < p^*(\beta_0)$. There are $\varepsilon > 0$ and $C < \infty$ such that*

$$\sup_{\beta \in [\beta_0, \beta_0 + \varepsilon]} \sup_{N \geq 0} \mathbb{E}[(W_N^\beta)^s] \leq C. \tag{3.1}$$

3.1. The common terminal field.

Fix real exponents

$$2 < s < p < q < p^*(\beta_0), \qquad a = \frac{s}{p} \in (0,1), \qquad r = \frac{q}{p} > 1. \tag{3.2}$$

For $j \geq 0$ and $x \in \mathbb{Z}^3$, let $W_N^{(j),x}$ denote the β_0 partition function using disorder times $j+1, \ldots, j+N$ and started at x. These martingales are uniformly bounded in L^q, and hence converge almost surely and in L^q [3, Theorem 4.4.6]. Write their limits as $X_x^{(j)}$. On a common probability-one set for the countable family (j,x), the six-term recursion passes to the limit:

$$X_x^{(j)} = \sum_y K(x,y) \eta_{j+1,y}^{\beta_0} X_y^{(j+1)}, \qquad \eta_{n,y}^\beta = e^{\beta \omega_{n,y} - \beta^2/2}, \tag{3.3}$$

where $K(x,y) = 1/6$ for nearest neighbors and zero otherwise. The whole field $X^{(j+1)}$ is independent of the first $j+1$ disorder slices and has the same joint law as $X^{(0)}$. Its coordinates have mean one. Put

$$C_p = \mathbb{E}[(X_0^{(0)})^p], \qquad C_q = \mathbb{E}[(X_0^{(0)})^q].$$

These finite marginal moments are unchanged by spatial translation.

We also need positive association. For bounded increasing functions f, g of one random variable U, with U' an independent copy,

$$\operatorname{Cov}(f(U), g(U)) = \tfrac{1}{2}\mathbb{E}[(f(U) - f(U'))(g(U) - g(U'))] \geq 0.$$

Conditioning on one coordinate and using induction proves the same inequality for increasing functions of finitely many independent coordinates. For a countable product, condition bounded f, g on the first m coordinates. Their conditional averages are increasing on the finite product and converge almost surely to f, g by the increasing-sigma-field convergence theorem [3, Theorem 4.6.8]. Bounded convergence passes the covariance inequality to the limit. Truncation and monotone convergence extend it to nonnegative functions with finite product expectation.

Each $X_x^{(j)}$ may be represented by the extended nonnegative liminf of its finite partition functions. This is increasing in every future disorder coordinate and agrees almost surely with the L^q limit. Thus the joint terminal field is associated. This construction uses its common future; its spatial coordinates are not replaced by independent copies.

Let μ be any finitely supported probability vector on $\mathbb{Z}^3$. To measure its terminal moment, define, using a copy X of the whole field $X^{(0)}$,

$$h(\mu) = \mathbb{E}_X\Big(\sum_x \mu_x X_x\Big)^p, \qquad H(\mu) = h(\mu)^a. \tag{3.4}$$

Jensen and mean one give the support-independent bounds

$$1 \le h(\mu) \le C_p, \qquad 1 \le H(\mu) \le C_p^a, \qquad \mathbb{E}_X\Big(\sum_x \mu_x X_x\Big)^q \le C_q. \tag{3.5}$$

On any fixed finite-support simplex, h and H are continuous by dominated convergence: the pth power of the sum of the terminal coordinates in that fixed support is an integrable dominator. In particular they are measurable at the finite endpoint distributions used below.

Predict the next endpoint law by $\alpha = \mu K$ and write

$$I = \sum_x \alpha_x^2 > 0, \qquad t_0 = \beta_0^2. \tag{3.6}$$

The positive support $J = \{x : \alpha_x > 0\}$ is finite. For independent standard Gaussians $(g_x)_{x\in J}$, independent of X, put

$$\eta_x(t) = e^{\sqrt{t}\, g_x - t/2}, \qquad A_t = \sum_x \alpha_x \eta_x(t), \qquad \nu_t(x) = \frac{\alpha_x \eta_x(t)}{A_t}. \tag{3.7}$$

Here t is the variance parameter β^2, not the semigroup time used in Section 2. For a positive vector $b = (b_x)_{x\in J}$ set

$$Z_b = \sum_x b_x X_x, \qquad B = \sum_x b_x, \qquad L(b) = \mathbb{E}_X Z_b^p, \qquad F(b) = L(b)^a. \tag{3.8}$$

The joint recursion (3.3) and homogeneity imply

$$\mathbb{E}_g L(\alpha\eta(t_0)) = h(\mu), \tag{3.9}$$

$$\mathbb{E}_{g,X}\Big(\sum_x \alpha_x \eta_x(t_0) X_x\Big)^q \le C_q, \tag{3.10}$$

$$F(\alpha\eta(t)) = A_t^s H(\nu_t). \tag{3.11}$$

Indeed, the sum in the first two lines has the law of $\sum_x \mu_x X_x^{(0)}$. This is where retaining the joint terminal field matters.

3.2. **Gaussian differentiation.** We first control the variation of $\mathbb{E}_g F(\alpha\eta(t))$ with t. For positive b, Jensen gives $B^p \le L(b) \le C_p B^p$. Differentiation under the X expectation gives

$$L_x = p\mathbb{E}_X[X_x Z_b^{p-1}], \qquad L_{xx} = p(p-1)\mathbb{E}_X[X_x^2 Z_b^{p-2}].$$

On compact subsets of the positive orthant, the integrand and these derivatives are bounded by an integrable multiple of $(\sum_{x\in J} X_x)^p$. Thus the differentiations are valid also for noninteger $p > 2$. Hölder gives

$$\mathbb{E}_X[X_x Z_b^{p-1}] \le C_p B^{p-1}, \qquad \mathbb{E}_X[X_x^2 Z_b^{p-2}] \le C_p B^{p-2},$$

and Cauchy–Schwarz gives $L_x^2 \le p^2 L \mathbb{E}_X[X_x^2 Z_b^{p-2}]$. Since $a < 1$,

$$F_{xx} = aL^{a-1}L_{xx} - a(1-a)L^{a-2}L_x^2 \le s(p-1)C_p B^{s-2}. \tag{3.12}$$

The same estimates also give

$$|F_x| \le sC_p B^{s-1}, \qquad |F_{xx}| \le \{s(p-1) + a(1-a)p^2\}C_p B^{s-2}. \tag{3.13}$$

Both $p-2$ and $s-2$ are positive.

Finite-dimensional Gaussian integration by parts now yields

$$\frac{d}{dt}\mathbb{E}_g F(\alpha\eta(t)) = \frac{1}{2}\mathbb{E}_g \sum_x (\alpha_x \eta_x(t))^2 F_{xx}(\alpha\eta(t)). \tag{3.14}$$

For completeness, differentiating before integration by parts gives $\sum_x b_x F_x(g_x/(2\sqrt{t}) - 1/2)$, with $b = \alpha\eta(t)$. The identity $\mathbb{E}_g(g_x f) = \mathbb{E}_g(\partial_{g_x} f)$ replaces $g_x b_x F_x$ by $\sqrt{t}(b_x F_x + b_x^2 F_{xx})$; the terms $b_x F_x$ cancel. On a compact interval of positive t, (3.13) bounds the relevant functions and derivatives by finite Gaussian exponentials with polynomial factors in g. These are integrable and dominate the difference quotients. They also make the boundary terms vanish against the Gaussian density. Dominated convergence and Fubini justify the derivative and integration by parts for each finite J.

By Hölder with exponents $s/2$ and $s/(s-2)$,

$$\mathbb{E}_g[\eta_x(t)^2 A_t^{s-2}] \le (\mathbb{E}\eta_x(t)^s)^{2/s} (\mathbb{E}A_t^s)^{(s-2)/s} \le e^{s(s-1)t/2}.$$

The last inequality follows from Jensen over the probability vector α. Consequently, (3.12)–(3.14) give

$$\frac{d}{dt}\mathbb{E}_g F(\alpha\eta(t)) \le \frac{s(p-1)C_p}{2} e^{s(s-1)t/2} I. \tag{3.15}$$

This bound has the same overlap I as the concavity estimate proved next.

3.3. **A uniform concavity estimate.** Recall $H(\mu) = h(\mu)^a$ and $F(b) = L(b)^a$. We show that $H(\mu)$ exceeds $\mathbb{E}_g F(\alpha\eta(t_0))$ by a fixed positive multiple of I, uniformly over all finite supports. Set

$$Y = L(\alpha\eta(t_0)), \qquad D = \frac{Y}{h(\mu)}.$$

By (3.9), D is a positive density of mean one. Since $r = q/p > 1$, conditional Jensen and (3.10) give

$$\mathbb{E}_g D^r \le \mathbb{E}_g Y^r \le \mathbb{E}_{g,X}\Big(\sum_x \alpha_x \eta_x(t_0) X_x\Big)^q \le C_q. \tag{3.16}$$

The scalar variable

$$G_\alpha = I^{-1/2} \sum_x \alpha_x g_x$$

is standard normal for every α. Let $r' = r/(r-1)$ and define

$$M = C_q^{1/r} (\mathbb{E}|G|^{2r'})^{1/r'}, \qquad G \sim N(0,1). \tag{3.17}$$

Hölder gives $\mathbb{E}_g[DG_\alpha^2] \le M$, with no dependence on J.

Association of the terminal field and Jensen, using $p - 1 > 1$, imply

$$\mathbb{E}_X[X_x Z_b^{p-1}] \ge \mathbb{E}_X X_x \, \mathbb{E}_X Z_b^{p-1} \ge B^{p-1}. \tag{3.18}$$

The product is integrable by the p-moment estimates above. Differentiating Y and integrating by parts in g_x therefore give

$$\begin{aligned}\mathbb{E}_g[G_\alpha Y] &= \frac{p\beta_0}{\sqrt{I}} \sum_x \alpha_x^2 \mathbb{E}_g\Big[\eta_x(t_0)\mathbb{E}_X(X_x Z_{\alpha\eta(t_0)}^{p-1})\Big] \\ &\ge p\beta_0\sqrt{I}.\end{aligned}$$

For the last inequality, (3.18) bounds the inner terminal expectation by $A_{t_0}^{p-1}$. Association in the fresh Gaussian slice and Jensen then yield $\mathbb{E}_g[\eta_x(t_0) A_{t_0}^{p-1}] \ge \mathbb{E}_g\eta_x(t_0)\, \mathbb{E}_g A_{t_0}^{p-1} \ge 1$. The growth estimates used in (3.14) also justify this integration by parts. Since $h(\mu) \le C_p$, we obtain

$$\mathbb{E}_g[DG_\alpha] \ge \frac{p\beta_0}{C_p}\sqrt{I}. \tag{3.19}$$

The original Gaussian mean of G_α is zero. Thus

$$\mathbb{E}_g[DG_\alpha] = \mathbb{E}_g[(\sqrt{D}-1)G_\alpha(\sqrt{D}+1)].$$

Cauchy–Schwarz, $(\sqrt{D}+1)^2 \le 2(D+1)$, and (3.17) imply

$$\mathbb{E}_g(\sqrt{D}-1)^2 \ge \frac{p^2\beta_0^2}{2C_p^2(1+M)}I. \tag{3.20}$$

Only the r moment of D is used; it need not have a second moment.

For $0 < a < 1$ and $z \ge 0$,

$$1 - a + az - z^a \ge \min(a, 1-a)(\sqrt{z}-1)^2. \tag{3.21}$$

If $a \le 1/2$, apply the tangent-line inequality to u^{2a} at $u = 1$. If $a \ge 1/2$, use the first case with $1-a$ and $1/z$, and multiply by z; the value at zero follows by continuity. Integrate (3.21) at D and use $\mathbb{E}_g D = 1$. By (3.9), $Y = h(\mu)D$, while $h(\mu)^a \ge 1$. Together with (3.20), this gives

$$H(\mu) - \mathbb{E}_g F(\alpha\eta(t_0)) \ge cI, \qquad c = \frac{\min(a,1-a)p^2\beta_0^2}{2C_p^2(1+M)} > 0. \tag{3.22}$$

3.4. **Perturbation and all-length iteration.** Keep β_0, s, p, q and the resulting H fixed. Define

$$Q = \frac{s(p-1)C_p}{2}e^{s(s-1)(t_0+1)/2}, \qquad \tau = \min\{1, c/(2Q)\} > 0.$$

Integrating (3.15) and using (3.11) and (3.22), for every $t \in [t_0, t_0+\tau]$ we get

$$\mathbb{E}_g[A_t^s H(\nu_t)] \le H(\mu) - cI + Q(t-t_0)I \le H(\mu). \tag{3.23}$$

The constants c, Q, τ are independent of μ and its support. Both the loss and the perturbation contain exactly $I = \sum_x \alpha_x^2$, so no lower bound on this overlap is required.

Fix β with $\beta^2 \in [t_0, t_0+\tau]$, and let

$$u_n(x) = E_0\left[\prod_{j=1}^n \eta_{j,S_j}^\beta;\ S_n = x\right], \qquad \mu_n(x) = \frac{u_n(x)}{W_n^\beta}.$$

These are the actual endpoint weights and endpoint law at the perturbed parameter. We have $\mu_0 = \delta_0$, $\alpha = \mu_n K$, and, conditionally on $\mathcal{F}_n$, the new slice is independent Gaussian. The ratio $W_{n+1}^\beta / W_n^\beta$ is A_{β^2} and the next endpoint law is ν_{β^2}. Equation (3.23) therefore says that

$$(W_n^\beta)^s H(\mu_n), \qquad n \ge 0,$$

is a nonnegative supermartingale. Every term is integrable by (2.1) and (3.5). Taking expectations and using $H \ge 1$ yields

$$\sup_{\beta^2 \in [t_0, t_0+\tau]} \sup_{N \ge 0} \mathbb{E}[(W_N^\beta)^s] \le H(\delta_0) = C_p^a. \tag{3.24}$$

Taking $\varepsilon = \sqrt{\beta_0^2 + \tau} - \beta_0$ proves Proposition 3.1.

3.5. **Proof of the moment-spectrum theorem.**

Proof of Theorem 1.1. By Proposition 2.1, p^* is finite and strictly decreasing on $(0, \beta_c]$, with limit infinity at zero and value $5/3$ at β_c. It remains to prove continuity. At every interior point with $p^*(\beta) > 2$, Proposition 3.1 applies to every real $s \in (2, p^*(\beta))$. Thus the right limit of the nonincreasing function p^* is at least each such s, and hence equals $p^*(\beta)$. At an interior point with $p^*(\beta) \le 2$, strict decrease and (2.4) give $p^*(\beta) > 5/3$; [4, Theorem 1.2(i)] supplies right continuity, including the case $p^*(\beta) = 2$.

Weak disorder holds at every $\beta \in (0, \beta_c]$, so (2.5) supplies left continuity throughout this domain, including the critical endpoint. The resulting continuous strictly decreasing function takes every finite value above $5/3$ by the intermediate value theorem: for such a value choose a sufficiently small positive b with $p^*(b)$ above it and use the interval $[b, \beta_c]$. Strict decrease makes the preimage unique. The value $5/3$ is attained only at β_c. □

AI assistance

OpenAI Codex was used extensively in developing the argument, checking sources and drafting the text. It also supplied internal reader and referee feedback. Independent mathematical review is requested.

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